

PATHIVADA RANJITH KUMAR

Vizianagaram, Andhra Pradesh

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PROFILE / CAREER OBJECTIVE

AI Engineer-focused Computer Science undergraduate (CGPA: 9.3/10) with hands-on experience in Machine Learning, Deep Learning, and LLM-based systems. Skilled in designing and deploying end-to-end AI solutions, including RAG pipelines, multi-agent systems, and scalable backend services using Python, PyTorch, FastAPI, LangChain, and vector databases. Strong foundation in NLP, sequence modeling, and model optimization, with a focus on building production-ready AI applications.

SKILLS & COMPETENCIES

- **Programming:** Python, Java, C
- **GenAI:** LLMs, RAG, Prompt Eng., Agentic AI, AI Agents, Multi-Agent Systems, Vector Search
- **AI/ML:** NLP, Deep Learning, Transformers, Embeddings, Semantic Search
- **Frameworks:** FastAPI, LangChain, LangGraph, Hugging Face, Streamlit
- **Databases:** MongoDB, Pinecone, FAISS, ChromaDB
- **Tools:** Git, GitHub, Docker, Linux, Postman, REST APIs
- **Core CS:** DSA, OOP, OS, CN, DBMS

EDUCATION

Rajiv Gandhi University of Knowledge Technologies, Nuzvid
B.Tech in Computer Science

2023 – 2027
CGPA: 9.3 / 10

Rajiv Gandhi University of Knowledge Technologies, Nuzvid
Pre-University Course (PUC) – MPC

2021 – 2023
CGPA: 10 / 10

PROJECTS

TriGuard AI — Multimodal Medical Triage Assistant (LangGraph + LLMs)

github.com/620593/TriGuard-AI-Medical-Triage-Assistant

- Developed a production-style multimodal medical triage platform integrating text, voice, OCR, X-ray, and skin image analysis using FastAPI, LangGraph, Groq LLMs, Whisper, Tesseract OCR, and React.
- Architected a stateful agentic AI pipeline with conditional routing, symptom extraction, risk scoring, emergency escalation, and LLM-based judge validation for reliable medical assistance.
- Implemented secure authentication (Google OAuth + JWT), MongoDB-backed conversational memory, and modular REST APIs, reducing average response latency by **30%** through workflow optimization.

AI-Powered Interview Preparation Assistant — Multi-Agent LLM System

github.com/620593/interview-prep

- Designed and developed a LangGraph-based multi-agent LLM system that generates personalized interview preparation roadmaps based on the target company, role, and preparation timeline.
- Built specialized AI agents for information extraction, company research, curriculum planning, coding pattern identification, study scheduling, and HTML dashboard generation.
- Developed a FastAPI backend with an interactive dashboard to present structured day-by-day interview plans and track preparation progress.

Handwritten Text Recognition using CRNN (CNN + BiLSTM + CTC)

github.com/620593/

- Designed and implemented a CRNN-based handwritten text recognition model combining CNN feature extraction, BiLSTM sequence modeling, and CTC loss.
- Improved recognition accuracy through transfer learning on synthetic datasets followed by fine-tuning on the IAM dataset, achieving **92% CCR**.
- Developed a complete OCR pipeline for line-level handwritten text recognition without requiring character segmentation.

Content-Based Image Retrieval (CBIR) Studio — ResNet-50, FAISS, Streamlit

github.com/620593/Content-Based-Image-Retrieval-CBIR-

- Engineered a Content-Based Image Retrieval (CBIR) system using pretrained ResNet-50 feature embeddings for semantic image similarity search.
- Implemented FAISS IndexFlatL2 vector indexing to retrieve visually similar images from a dataset of approximately 25K images with low-latency nearest-neighbor search.
- Built an end-to-end retrieval pipeline integrating feature extraction, vector indexing, metadata mapping, and an interactive Streamlit interface for real-time image search.

LEADERSHIP & ACTIVITIES

Event Coordinator — Teckzite 2K25, RGUKT Nuzvid

Dec 2025

- Built an event management dashboard and coordinated 8+ technical events.

Web Designer — CYGNUS 2K25, RGUKT Nuzvid

Mar 2025

ACHIEVEMENTS

- **NPTEL Artificial Intelligence** — Scored in Top 5% nationally among all participants.
- **NPTEL Cloud Computing** — Ranked in Top 2% nationally among all participants.
- **NPTEL Deep Learning** — Elite + Silver Certificate
- **GATE (2026)** — Score: 697 (AIR 1202).